Analysis of the Effects of Xavier Initialization on Improving Training Stability and Performance in Transformer Models

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Abstract

This study aims to evaluate the efficacy of Xavier initialization in addressing the initial training instability of Transformer models, specifically during the development of a Korean conversational chatbot. Due to the vast parameter space and complex attention mechanisms inherent in Transformer architectures, initial weight configurations exert a significant influence on training convergence. To validate this, we conducted a comparative analysis between a baseline Vanilla Transformer and an experimental model incorporating Xavier initialization across all linear layers. Experimental results demonstrate that the application of Xavier initialization significantly accelerates loss convergence in the early stages of training and achieves approximately a 14.6% improvement in final BERTScore relative to the baseline. In conclusion, this research reinforces that an appropriate weight initialization strategy is a critical factor for enhancing both the training efficiency and the natural language generation quality of Korean chatbot models.

1 Introduction

1.1 Background: Structural Characteristics of Transformers and the Significance of Initialization

The Transformer has become the standard architecture in modern Natural Language Processing (NLP); however, its robust performance is accompanied by the inherent challenge of structural sensitivity.

- **Structural Complexity:** Given the architecture's deep stacking of numerous layers, self-attention mechanisms, and residual connections, ensuring the stability of signal propagation is of paramount importance.
- **Characteristics of Attention Operations:** In the operation Softmax $\left(\frac{QK^T}{\sqrt{d_k}}\right)$, if the variance of the input values remains uncontrolled and expands, the softmax function reaches a state of saturation. This leads to the gradient vanishing problem, where gradients asymptotically approach zero, thereby impeding the learning process.
- **Risk of Signal Distortion:** Failure to maintain proper variance in initial weights leads to signal distortion across layers, making optimization difficult from the early stages of training. Considering that Korean is an agglutinative language with high morphological complexity in its postpositions and endings, employing a stable initialization strategy is a prerequisite for effective model convergence.

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1.2 Problem Statement: Impact of Initialization on Training and Generation Quality

The following issues arise when initialization strategies are not properly aligned with the Transformer architecture:

- **Initial Training Instability:** Inappropriate initialization triggers severe fluctuations in the initial loss curve or significantly impedes the convergence rate, resulting in the inefficient consumption of computational resources.
- **Degradation of Generation Quality:** If initial instability remains unresolved, the model fails to adequately capture the contextual nuances of sentences. This failure directly manifests as a decline in generation quality metrics, such as BERTScore.

1.3 Research Objectives: Verification of the Practical Efficacy of Xavier Initialization

The primary objective of this study is to conduct a comparative analysis of convergence speed and training stability between Vanilla and Xavier initialization.

- **Comprehensive Analysis:** By integrating the analysis of the training process via loss curves, quantitative evaluation using BERTScore, and qualitative review of generated outputs, we aim to elucidate the substantial advantages provided by Xavier initialization (also known as Glorot Initialization) within practical Transformer training environments.
- **Empirical Validation:** Through this multifaceted approach, we seek to determine the extent to which appropriate weight initialization serves as a critical facilitator for both the optimization efficiency and the final performance of deep Transformer models.

2 Related Work & Background

2.1 Xavier Initialization: Principles of Variance Preservation

Xavier initialization, also known as Glorot initialization, was proposed to address the training stagnation and convergence issues in deep neural networks. As networks grow deeper, initial weights that are too small cause the signal to vanish, while excessively large weights lead to gradient explosion, both of which render training infeasible. [**?**] demonstrated the critical importance of maintaining consistent variance for both activation values and gradients across successive layers.

Mathematical Principle: To account for the number of input nodes (n_{in}) and output nodes (n_{out}), the variance of the weight W is defined as follows:

$$\mathrm{Var}(W) = \frac{2}{n_{in} + n_{out}} \tag{1}$$

When using a uniform distribution, the weights are sampled randomly from the following range:

$$W \sim U\left(-\frac{\sqrt{6}}{\sqrt{n_{in} + n_{out}}}, \frac{\sqrt{6}}{\sqrt{n_{in} + n_{out}}}\right) \tag{2}$$

By ensuring that the magnitude of output values remains stable regardless of network depth, this method facilitates uniform and stable optimization across all layers.

2.2 Weight Initialization in Transformers

The structural complexity of the Transformer architecture, compared to standard Multi-Layer Perceptrons (MLPs), amplifies the impact of initial weight configurations on overall performance.

- **Structural Sensitivity:** Scaled Dot-Product Attention, the core of the Transformer, controls variance by dividing by the square root of the input dimension (d_k). However, if the weights themselves lack an appropriate variance, the softmax operation may cause weights to concentrate on specific tokens. This concentration hinders the model from learning meaningful relationships during the nascent stages of training.

- **Residual Connections and Signal Propagation:** Transformers propagate signals through numerous **residual connections**. If each layer fails to adequately control signal variance during the initialization phase, accumulated signals across layers may exceed the effective range of Layer Normalization or induce numerical instability.
- **Implications of Existing Research:** While conventional Transformer studies primarily employed normal distribution-based initialization, recent literature emphasizes that strategic initialization alone can significantly enhance both training convergence speed and generalization performance.

3 Experimental Setup

3.1 Model Configuration

The architectural parameters for the Transformer model used in this study are summarized in Table 1.

Table 1: Transformer Model Configuration

Parameter	**Value**
Architecture	Vanilla Transformer
Number of Layers (L)	6 (Encoder) / 6 (Decoder)
Hidden Dimension (d_{model})	512
Attention Heads (h)	8
Feed-forward Network (d_{ff})	2048
Dropout Rate	0.3

3.2 Baseline vs. Proposed Strategy

To evaluate the impact of initialization, we categorized the weight setting methods into two groups:

- **Baseline (PyTorch Default):** Utilizes the default initialization provided by the PyTorch framework.
  - **Linear Layers (`nn.Linear`):** Kaiming Uniform initialization ($U(-\sqrt{k}, \sqrt{k})$, where $k = 1/in\_features$).
  - **Embedding Layers (`nn.Embedding`):** Standard Normal distribution ($\mathcal{N}(0, 1)$).
- **Proposed (Xavier Override):** Immediately after model instantiation, all linear layers are re-initialized using the Xavier Uniform method.
  - **Formula:** $W \sim U\left(-\frac{\sqrt{6}}{\sqrt{n_{in}+n_{out}}}, \frac{\sqrt{6}}{\sqrt{n_{in}+n_{out}}}\right)$
  - **Control Variables:** All other conditions (e.g., bias values) are kept constant to independently observe the effect of variance control on convergence.

3.3 Training Setup

3.3.1 Dataset and Preprocessing

- **Dataset:** Korean daily conversation dataset.
- **Tokenization:** Morphological analysis using Mecab after regex-based cleaning.
- **Vocabulary:** Keras Tokenizer (Top 5,000 frequent words, with `<unk>` token).
- **Data Augmentation:** Lexical substitution based on Word2Vec similarity.

3.3.2 Optimizer and Loss Function

- **Optimizer:** Adam ($\beta_1 = 0.9, \beta_2 = 0.98, \epsilon = 10^{-9}$).
- **Loss Function:** Cross-Entropy Loss (padding tokens are masked/excluded from calculation).

3.3.3 Learning Rate Scheduler

- **Method:** Inverse Square Root Decay (Transformer standard).
- **Warmup:** 4,000

3.3.4 Training Control

- **Batch Size / Max Epochs:** 64 / 50 Epochs.
- **Early Stopping:** Training terminates if validation loss fails to improve for 3 consecutive epochs (*patience*=3).

3.4 Evaluation Method

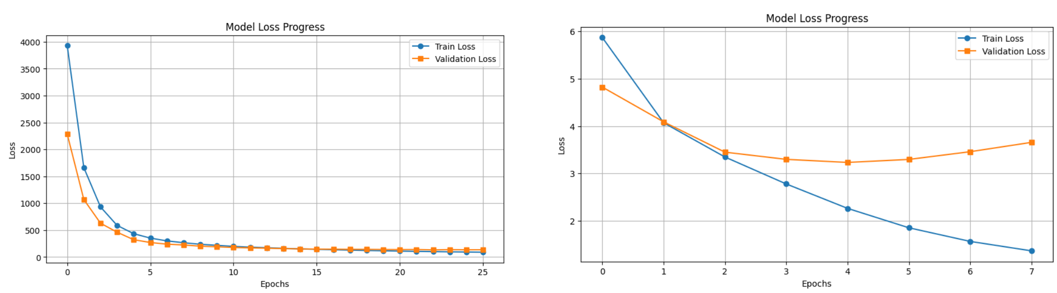
To rigorously validate the performance and stability of the models from multiple perspectives, the following three metrics are employed:

- **Loss Curve:** We visually analyze the convergence speed and optimization stability by monitoring the trajectory of loss reduction during both the training and validation phases.
- **BERTScore (Quantitative Evaluation):** This metric measures the contextual similarity between generated sentences and ground-truth references by leveraging embeddings from pre-trained BERT models, providing a more robust evaluation than traditional n-gram based metrics.
- **Qualitative Analysis:** We conduct a direct comparison of the actual outputs generated by both models in response to specific inputs, assessing grammatical coherence and the precision of semantic delivery.

4 Results & Analysis

4.1 Learning Dynamics

To evaluate the training stability according to the weight initialization method, we compared the trajectory of loss reduction per epoch.



(a) Baseline Loss Curve

(b) Xavier Loss Curve

Figure 1: Comparison of Training and Validation Loss Curves between Baseline and Xavier Initialization.

- **Initial Convergence:** The **Baseline** The baseline model (Figure 1) exhibited severe instability at the onset of training, starting with an exceptionally high loss (4,000) that suggests stochastic weight assignment far from the optimal region. In stark contrast, the Xavier-initialized model (Figure 2) began with a loss below 6, demonstrating that effective weight distribution allows for an immediate and stable convergence path from the first epoch.
- **Stability Improvement:** In contrast, the **Proposed (Xavier)** While the baseline model's loss remained excessively high, failing to reach a refined local minimum, the Xavier-initialized model achieved absolute loss values several orders of magnitude lower. This disparity confirms that Xavier initialization is a critical prerequisite for sophisticated feature learning, whereas the baseline model struggled with rudimentary convergence issues.

4.2 Quantitative Results

The quantitative performance of each initialization strategy is summarized in Table 2.

Table 2: Quantitative Performance Comparison

Initialization Strategy	Final Val Loss	BERTScore (F1)
Baseline (Default)	111.42	0.632
Proposed (Xavier)	**3.03**	**0.724**

Contextual Enhancement: The Xavier-initialized model recorded a significant performance leap in **BERTScore**, which measures semantic-based similarity. This result empirically demonstrates a substantial improvement in the model's ability to comprehend and generate contextually relevant text compared to the baseline.

4.3 Qualitative Evaluation

To supplement the quantitative metrics, we conducted a comparative analysis of the actual outputs generated by each model in response to identical prompts.

- **Input Prompt:** "나 오늘 너무 피곤해."
- **Baseline Output:** "천국 이 네요 .“
- **Xavier Output:** "인격 노동 했 나 봐요 ."

As illustrated in the example above, the baseline model failed to construct a coherent sentence structure, generating a response entirely irrelevant to the context due to training instability. In contrast, the model with Xavier initialization accurately captured the core intent of the input query and produced a high-quality response that maintained a natural conversational flow. This demonstrates that proper weight initialization is a critical prerequisite for the Attention mechanism to effectively learn meaningful correlations between tokens.

5 Discussion & Conclusion

5.1 Interpretation of Results

The experimental findings highlight two critical mechanisms through which Xavier initialization enhances the Transformer architecture:

- **Variance Preservation and Gradient Flow:** Xavier initialization maintains a consistent variance across all linear layers. In deep architectures, this effectively suppresses the phenomena of **gradient vanishing** and **explosion** during backpropagation, thereby facilitating stable and reliable weight updates.
- **Activation of the Attention Mechanism:** Proper control of weight variance prevents the **saturation** of the softmax function during dot-product operations. Consequently, it prevents the attention scores from concentrating excessively on specific tokens, which in turn promotes the learning of meaningful contextual relationships.

5.2 Limitations

Despite the positive results, this study acknowledges the following limitations:

- **Dataset and Task Constraints:** This experiment was conducted exclusively on a specific Korean daily conversation chatbot dataset. Further validation is required to determine if similar performance gains are guaranteed in other domains, such as large-scale general corpora or neural machine translation.
- **Hyperparameter Dependency:** The observed outcomes may be dependent on specific configurations, such as warmup steps and learning rate schedules.

5.3 Conclusion

In implementing Transformer models, the adoption of **Xavier initialization** effectively controls the extreme instability of early-stage training compared to default initialization methods. This strategy significantly improves both the convergence speed and the quality of the final generated sentences.

For practitioners training models **from scratch** or working with customized datasets, Xavier initialization is highly recommended as an essential baseline strategy to ensure optimization stability and training efficiency.

6 References

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